

Bridging measurements and simulations: A data assimilation approach to real time analysis

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Abstract

Accurate reconstruction of wall-bounded flows from sparse measurements depends not only on the assimilation method and measured variables, but also critically on the spatial distribution of sensors. While previous data assimilation (DA) studies have successfully combined numerical simulations with experimental observations to improve reconstruction accuracy (Hayase & Hayashi, 1997; Neeteson & Rival, 2020), the question of where sensors should be placed remains central. Optimal sensor positioning has been shown to significantly enhance estimation quality while reducing instrumentation requirements (Mons & Marquet, 2021).

In this work, we extend our previous analysis of variable selection by investigating the influence of sensor placement along the surface of a flat plate undergoing a sudden acceleration boundary-layer transition. High-resolution simulations provide the reference dataset, from which synthetic wall observations (temperature, pressure, shear stress) are sampled according to different spatial distributions. Strategies tested include uniform spacing, localized clustering near flow features, and adaptive placement based on wall-gradient indicators. These observations are assimilated into coarse-resolution simulations using lightweight DA schemes such as nudging and Kalman-based filters.

Performance is assessed in terms of reconstruction accuracy (RMSE), assimilation cost, and robustness to sparse configurations. Results show that sensors placed near the leading edge and in regions of strong shear provide the highest gains in reconstruction quality. Uniform distributions yield balanced improvements, while adaptive gradient-based placement achieves the best trade-off between accuracy and sensor economy.

These findings highlight the importance of optimized sensor placement for DA in wall-bounded flows, providing guidelines for integrating surface-based measurements into computational fluid dynamics workflows with minimal instrumentation requirements.